

# LOCALIZATION IN THE COMMUTANT, STAGE 9:

the mean of a random ReLU network as a central localization,  
with the exact slice transport, the collective-mode sensitivity law  
and the capacity of input tilts

Stage 9 of the quantum-cut / noncommutative-geometry programme

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## Abstract

We apply the stage-8 principle (a localization process is a positive martingale in the commutant, whose central part is the transverse measure on the leaf space) to the WhetBench problem: the per-neuron mean  $\mathbb{E} h_L$  of a bias-free He-initialised ReLU multilayer perceptron with standard Gaussian input,  $n = 1024$ ,  $L = 16$ . Results. (A) A defect-path theorem: the error of any estimator exact on point masses is minus its expected drift along any complete localization of the input, i.e. its second variation against the scheme's driving covariance; on Eldan's path this is the Euler–Stein defect  $\delta = e - \Delta e$ , and for the Gaussian closure at full scale the exact  $\delta$  explains 89% of the error ( $4.1 \cdot 10^{-6} \rightarrow 4.4 \cdot 10^{-7}$ ), with the parameter-free midpoint  $(e + \Delta e)/2$  nearly as good, and for the organisers' exact first-order chain it explains 81% ( $3.5 \cdot 10^{-8} \rightarrow 6.6 \cdot 10^{-9}$ , below the leaders' raw accuracy). (B) Input tilts have no capacity:  $\delta$  is a 2% residual of an  $O(1)$  identity, random probes need  $10^9$  samples, and only the  $2n + 1$  coordinate sum resolves it. (C) The index-shift lemma  $d'_k = d_{k+1}$  and the Gram theorem  $\delta = -\langle \nabla^2 \mathcal{G}, \Gamma \rangle$ : births, arms, slices and gate covariances are the derivatives of the Gaussian moment map, and the organisers' 36-diagram first-order chain is this calculus. We port that chain to numpy, match it to  $10^{-15}$ , and find its accuracy ( $3.5 \cdot 10^{-8}$ ) rests on two things: the first-order term table and one radial  $\kappa_4$  scalar per layer, the scale mixture of the collective mode (a factor 16); our own chain had the first and lacked the second. (D) Its cost is the memory of  $\kappa_3$ :  $12n^3$  per source-age,  $0.5\text{--}0.7B$  in float32 under the grader's measured rules, against a floor of  $0.1B$ ; the gated transport cocycle has slowly decaying rank but the slices are Hadamard products of the legs, so Frobenius compression of young sources fails (a factor 26 at  $k = 128$ ) while a four-age window with a 384-dimensional shared basis is lossless and still costs  $0.5B$ ; the old content is a new shape at every layer (correlation 0.3–0.5 with the young content), so it cannot be regenerated either. Every identity is checked by the accompanying scripts.

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# 1 The problem and the principle

## 1.1 The estimation problem

Fix weights  $W_1, \dots, W_L \in \mathbb{R}^{n \times n}$ , drawn once from  $N(0, 2/n)$  (He initialisation) and then held fixed. For an input  $x \in \mathbb{R}^n$  put  $h_0 = x$ ,  $z_l = W_l h_{l-1}$ ,  $h_l = \text{relu}(z_l)$ . The target is the vector of post-activation means

$$u = \mathbb{E}_{x \sim N(0, I_n)} h_L(x) \in \mathbb{R}^n,$$

scored by the mean squared error over the  $n$  neurons, multiplied by  $\max(0.1, C/B)$  with  $C$  the metered floating point count and  $B = 2^{41}$ . The map  $F = h_L$  is positively homogeneous of degree one and piecewise linear. The reference answer is a Monte Carlo average over  $10^9$  inputs, whose per-neuron noise ( $\sim 10^{-5}$ ) is below every error discussed here.

## 1.2 Estimators are functions on the state space; exact ones are affine

Let  $\mathcal{S}$  be the set of probability laws  $\nu$  on  $\mathbb{R}^n$  with enough moments. Every deterministic estimator we consider is a map  $E : \mathcal{S}_0 \rightarrow \mathbb{R}^n$  defined on a subfamily  $\mathcal{S}_0 \subseteq \mathcal{S}$  that contains the point masses  $\delta_x$  and the Gaussians  $N(m, \Sigma)$ ; a cumulant chain is defined on Gaussians by running its recursion from the given  $(m, \Sigma)$ , and on point masses by the forward pass,  $E(\delta_x) = F(x)$ . The exact answer

$$u(\nu) = \nu(F) = \int F d\nu$$

is *affine* on  $\mathcal{S}$ . This is the whole content of exactness: in the commutant language of [1],  $u$  is the pairing of the state  $\nu$  with the observable  $F$ , and a localization process of  $\nu$  (a positive martingale of Radon–Nikodym derivatives in the commutant) turns  $u$  into a martingale. A chain is not affine: it is exact on point masses and wrong on mixtures, and the gap between the two is what we have to compute.

**Definition 1.1** (Localization process). A localization process of  $\nu$  is a sequence or continuous family  $(\nu_t)_{t \geq 0}$  of random laws with  $\nu_0 = \nu$ , adapted to a filtration  $(\mathcal{F}_t)$ , such that  $\mathbb{E}[\nu_{t+s}(g) \mid \mathcal{F}_t] = \nu_t(g)$  for every bounded  $g$ , and *complete* if  $\nu_t \rightarrow \delta_X$  almost surely for a random  $X \sim \nu$  [4, 1]. Three schemes are used below: Eldan’s isotropic tilt,  $\nu_t = \text{Law}(x \mid y_t)$  with  $y_t = tX + B_t$ , so that for  $\nu = N(0, I)$  one has  $\nu_t = N(m_t, I/(1+t))$  with  $m_t$  a martingale of quadratic variation  $d\langle m \rangle_t = (1+t)^{-2} I dt$ ; the *directional* tilt, the same observation with  $y_t = t\langle v, X \rangle + B_t$  for a unit vector  $v$ , which leaves  $\nu_t = N(m_t, I - \frac{t}{1+t} vv^\top)$ ; and pinning, which reveals the coordinates of  $X$  one at a time.

**Theorem 1.2** (Defect–path formula). *Let  $E : \mathcal{S}_0 \rightarrow \mathbb{R}^n$  be twice continuously Fréchet differentiable along the Gaussian family and exact on point masses, and let  $(\nu_t)$  be a complete localization process of  $\nu \in \mathcal{S}_0$  whose components stay in  $\mathcal{S}_0$  and have bounded quadratic variation. Then*

$$u(\nu) - E(\nu) = -\mathbb{E} \int_0^\infty \frac{1}{2} E''(\nu_t) [d\langle \nu \rangle_t] = -\mathbb{E} \int_0^\infty \text{drift}_t(E(\nu_t)) dt. \quad (1)$$

*In words: the error of the estimator at  $\nu$  is minus the expected total drift of the estimator along any complete localization of  $\nu$ , and that drift is the second variation of  $E$  in the directions the scheme localizes.*

*Proof.*  $u(\nu_t)$  is a martingale (the law of iterated expectations, or affinity of  $u$  plus the martingale property of  $\nu_t$ ), so  $u(\nu) = \mathbb{E} u(\nu_\infty) = \mathbb{E} F(X)$ . By Itô’s formula for the semimartingale  $\nu_t$  in the parametrisation of  $\mathcal{S}_0$ ,  $dE(\nu_t) = E'(\nu_t)[d\nu_t] + \frac{1}{2} E''(\nu_t)[d\langle \nu \rangle_t]$ , and the first term has zero mean. Hence  $\mathbb{E} E(\nu_\infty) - E(\nu) = \mathbb{E} \int_0^\infty \frac{1}{2} E''(\nu_t)[d\langle \nu \rangle_t]$ . Exactness on point masses gives  $E(\nu_\infty) = F(X)$  almost surely, so  $\mathbb{E} E(\nu_\infty) = u(\nu)$ . For a discrete scheme the same identity holds with the drift replaced by the conditional Jensen gaps  $\mathbb{E}[E(\nu_{t+1}) \mid \mathcal{F}_t] - E(\nu_t)$ .  $\square$

**Corollary 1.3** (Eldan’s path for a homogeneous network). *Write  $e(y) = E(N(y, I))$  for a chain  $E$  that is exact on point masses and transforms like  $F$  under scaling,  $E(N(m, sI)) = \sqrt{s} e(m/\sqrt{s})$ . Then with  $\delta(y) = e(y) - y \cdot \nabla e(y) - \Delta e(y)$ ,*

$$u - e(0) = -\frac{1}{2} \int_0^\infty (1+\tau)^{-3/2} \mathbb{E}_{y \sim N(0, \tau I)} \delta(y) d\tau, \quad (2)$$

*and the weight  $\frac{1}{2}(1+\tau)^{-3/2} d\tau$  is a probability measure on  $[0, \infty)$ . At the origin  $\delta(0) = e(0) - \Delta e(0)$ , and the exact answer satisfies  $u = \Delta_y u(y)|_{y=0}$  (Euler–Stein).*

*Proof.* On Eldan’s path  $\nu_t = N(m_t, s_t I)$ ,  $s_t = 1/(1+t)$ ,  $ds_t = -s_t^2 dt$ ,  $d\langle m \rangle_t = s_t^2 I dt$ . With  $E(m, sI) = \sqrt{s} e(m/\sqrt{s})$  one finds  $\partial_s E = \frac{1}{2\sqrt{s}}(e - y \cdot \nabla e)$  and  $\frac{1}{2}\Delta_m E = \frac{1}{2\sqrt{s}}\Delta e$  at  $y = m/\sqrt{s}$ , so the drift is  $-s^2(\partial_s E - \frac{1}{2}\Delta_m E) = -\frac{1}{2}s^{3/2}\delta(y)$ . Since  $y_t = m_t\sqrt{1+t} \sim N(0, tI)$ , Theorem 1.2 gives (2) with  $\tau = t$ . For the exact answer,  $u(y) = \mathbb{E}F(y+Z)$  obeys the heat equation  $\partial_s u = \frac{1}{2}\Delta_m u$ , and for a 1-homogeneous  $F$  Euler’s relation  $x \cdot \nabla F = F$  and Stein’s lemma  $\mathbb{E}[x \cdot \nabla F(x)] = \mathbb{E} \Delta F(x)$  give  $u = \Delta_y u|_0$ .  $\square$

This corollary is the heat-flow defect of the parallel note (its “heat defect” is  $D = \partial_\Sigma E - \frac{1}{2}\nabla_m^2 E$ ; here  $\frac{1}{2}s^{3/2}\delta$  is the trace of  $D\Sigma_t^2$  on the isotropic path). Theorem 1.2 is the scheme-independent statement behind it, and the choice of scheme is where the stage-8 picture adds something.

**Corollary 1.4** (Directional and pinning schemes). *(i) For the directional tilt along a unit vector  $v$ , the path is  $\nu_t = N(m_t, I - \frac{t}{1+t}vv^\top)$  with  $d\langle m \rangle_t = (1+t)^{-2}vv^\top dt$ , the endpoint is  $\nu_\infty = N(\langle v, X \rangle v, I - vv^\top)$ , and*

$$\mathbb{E}_y E(N(yv, I - vv^\top)) - E(N(0, I)) = - \int_0^\infty (1+t)^{-2} \mathbb{E} \left[ \frac{1}{2} \partial_v^2 E - \partial_{\Sigma_{vv}} E \right] (\nu_t) dt,$$

*the left side being the error of the  $v$ -localized estimator minus the error of  $E$ . So the drift along  $v$  measures the difference between the chain and its  $v$ -localization, and the chain’s own error is recovered from directional drifts only by summing over an orthonormal basis of directions—which is Eldan’s path. (ii) For pinning (reveal  $X_i$  in order), the Jensen gap at step  $i$  is  $\mathbb{E}_{X_i} E(\nu_{i-1} | X_i) - E(\nu_{i-1})$ , a one-dimensional Gaussian integral of the chain run on the conditioned law; the chain’s error is minus the sum of these  $n$  gaps.*

**Remark 1.5** (Which errors a scheme sees). The second variation  $E''(\nu)[\dot{\nu}, \dot{\nu}]$  is a quadratic form on the tangent space of  $\mathcal{S}_0$  at  $\nu$ . A scheme supplies a positive form  $d\langle \nu \rangle$  (its driving covariance  $C_t$  in the language of [1, Thm. 3.3], which also fixes the trickle-down term  $KC_tK$  of that theorem). Eldan’s path drives every mean direction equally and the covariance only through the scale; the directional tilt drives one direction; pinning drives the coordinate directions and their conditional variances. Which scheme probes a given chain cheaply is therefore a question about the *spectrum* of the chain’s second variation on the input side. Section 3 measures it: for the chains of this note the excess second variation  $\partial_v^2 e - \Delta e/n$  over unit input directions  $v$  has a flat spectrum (the top direction carries 5% of it at  $n = 256$ , and is not the coherent direction  $W_1^\top \mathbf{1}$ ), and single directional second differences are nearly uncorrelated with the error. The chain’s error is an isotropic, trace-like object on the input side—the Laplacian of Corollary 1.3—and the only cheap way to get at it is not a tilt of the input but a second-order expansion *inside* the chain (Section 4). The collective mode that the chain mishandles lives in the gate algebra of the hidden layers, not in the input, which is the content of the capacity bound of Section 3.

## 2 Births are linear-tilt responses of the gate algebra

We now compute the second variation of a chain explicitly. The result is that the objects a cumulant chain transports (births at the kinks, gated arms,  $(2, 1)$  slices) are the first and second derivatives of the Gaussian moment map under a shift of the mean, and that the Euler–Stein defect of Section 3 is the curvature of the downstream chain contracted with the Gram matrix of those derivatives.

## 2.1 The chaos index shifts under a tilt

For  $u \sim N(0, 1)$  and  $a \in \mathbb{R}$  let  $r_a(u) = \text{relu}(u+a)$  with Hermite coefficients  $d_k(a) = \mathbb{E}[r_a(u)\text{He}_k(u)]$ :

$$d_0 = \varphi(a) + a\Phi(a), \quad d_1 = \Phi(a), \quad d_k = \varphi(a)\text{He}_{k-2}(-a) \quad (k \geq 2),$$

so that for standardised  $(u, v)$  with correlation  $\rho$ ,  $\mathbb{E}[r_a(u)r_b(v)] = \sum_{k \geq 0} \rho^k d_k(a)d_k(b)/k!$ .

**Lemma 2.1** (Index shift).  $d'_k(a) = d_{k+1}(a)$  for every  $k \geq 0$ .

*Proof.*  $d'_0 = -a\varphi + \Phi + a\varphi = \Phi = d_1$  and  $d'_1 = \varphi = d_2$ . For  $k \geq 2$ , with  $x = -a$ ,  $\frac{d}{da}[\varphi(a)\text{He}_{k-2}(x)] = \varphi[x\text{He}_{k-2}(x) - (k-2)\text{He}_{k-3}(x)] = \varphi\text{He}_{k-1}(x) = d_{k+1}(a)$  by  $\text{He}'_m = m\text{He}_{m-1}$  and the three-term recursion  $\text{He}_{m+1}(x) = x\text{He}_m(x) - m\text{He}_{m-1}(x)$ .  $\square$

(Checked numerically to  $2 \cdot 10^{-8}$  for  $k \leq 10$  in `verify_stage9.py`.) The lemma is Stein's identity  $\mathbb{E}[f'(u)\text{He}_k(u)] = \mathbb{E}[f(u)\text{He}_{k+1}(u)]$  for  $f = r_a$ ; what matters here is that the tilt  $a \mapsto a + t$  of the Gaussian acts on the chaos expansion of the RELU by raising the index.

Let now  $z \sim N(m, S)$  be the pre-activation of one layer,  $\sigma_a = \sqrt{S_{aa}}$ ,  $\alpha_a = m_a/\sigma_a$ ,  $\rho_{ab} = S_{ab}/\sigma_a\sigma_b$ , and

$$\mu_a = \sigma_a d_0(\alpha_a), \quad M_{ab} = \mathbb{E}[\text{relu } z_a \text{ relu } z_b] = \sigma_a \sigma_b \sum_k \rho_{ab}^k d_k(\alpha_a) d_k(\alpha_b) / k!$$

the post-activation mean and raw second moment. Lemma 2.1 gives every derivative of the closure:

$$\partial_{m_a} \mu_a = \Phi(\alpha_a) =: g_a, \quad \partial_{m_a}^2 \mu_a = \varphi(\alpha_a) / \sigma_a =: w_a, \quad (3)$$

$$\partial_{m_a} M_{ab} = \sigma_b \sum_k \rho^k d_{k+1}(\alpha_a) d_k(\alpha_b) / k! = \mathbb{E}[\theta(z_a) \text{relu } z_b] =: T_{ab}, \quad (4)$$

$$\partial_{S_{ab}} M_{ab} = \sum_k \rho^k d_{k+1}(\alpha_a) d_{k+1}(\alpha_b) / k! = \mathbb{E}[\theta(z_a) \theta(z_b)] = \partial_{m_a} \partial_{m_b} M_{ab} =: G_{ab}, \quad (5)$$

$$\partial_{m_a}^2 M_{ab} = \frac{\sigma_b}{\sigma_a} \sum_k \rho^k d_{k+2}(\alpha_a) d_k(\alpha_b) / k! = w_a \mu_b + \frac{\sigma_b}{\sigma_a} \sum_{k \geq 1} \rho^k d_{k+2}(\alpha_a) d_k(\alpha_b) / k!. \quad (6)$$

Here  $\theta$  is the Heaviside gate. The identities with expectations are Price's theorem  $\partial_{S_{ab}} \mathbb{E}[f(z_a)h(z_b)] = \mathbb{E}[f'(z_a)h'(z_b)]$  and Stein's lemma; the closure is exact for one layer because it satisfies the heat equation  $\partial_{S_{ab}} = \partial_{m_a} \partial_{m_b}$  in its own parameters. The dictionary with the cumulant chain of [2] and of Section 5 is now literal:  $w_a = \varphi(\alpha_a)/\sigma_a$  is the *birth* strength at the kink of neuron  $a$ , the second derivative of the mean under a tilt;  $g_a = \Phi(\alpha_a)$  gates the arms;  $T_{ab}$  is the exact  $(2, 1)$  slice function  $\mathbb{E}[\theta(z_a) \text{relu } z_b]$  through which a tilt at  $a$  moves the covariance row of  $a$ ;  $G_{ab}$  is the gate covariance  $\mathbb{E}[\theta_a \theta_b]$  that transports a covariance perturbation. Births are the curvature of the tilt response; arms and slices are its gradient.

## 2.2 The defect is downstream curvature against the tilt Gram

Write the chain as  $e(y) = \mathcal{G}(\theta(y))$ , where  $\theta(y) = (\mu(y), M(y))$  are the exact first-layer post-activation moments for the input law  $N(y, I)$  (exact because  $z_1 = W_1 x$  is Gaussian) and  $\mathcal{G}$  is the chain from layer 2 on, a function of the Gaussian surrogate  $N(\mu, M - \mu\mu^\top)$  of  $h_1$ .

**Theorem 2.2** (Defect = curvature  $\times$  Gram). *Let  $\mathcal{G}$  be homogeneous,  $\mathcal{G}(\lambda\mu, \lambda^2 M) = \lambda \mathcal{G}(\mu, M)$ , and twice differentiable. Then for every  $y$ ,*

$$\delta(y) = e - y \cdot \nabla e - \Delta e = - \sum_{A,B} \partial_A \partial_B \mathcal{G}(\theta(y)) \Gamma_{AB}(y), \quad \Gamma_{AB}(y) = \langle \nabla_y \theta_A(y), \nabla_y \theta_B(y) \rangle, \quad (7)$$

the sum running over the moment indices  $A, B \in \{\mu_a\} \cup \{M_{ab}\}$ . With  $S = W_1 W_1^\top$  and the rows  $w_a$  of  $W_1$ , at  $y = 0$  the Gram matrix is

$$\Gamma_{\mu_a \mu_b} = g_a g_b S_{ab}, \quad \Gamma_{\mu_a M_{bc}} = g_a (T_{bc} S_{ab} + T_{cb} S_{ac}), \quad \Gamma_{M_{ab} M_{cd}} = T_{ab} T_{cd} S_{ac} + T_{ab} T_{dc} S_{ad} + T_{ba} T_{cd} S_{bc} + T_{ba} T_{dc} S_{bd}.$$

Consequently  $u - e(0) = \int_0^\infty w(\tau) \mathbb{E}_{y \sim N(0, \tau I)} \langle \nabla^2 \mathcal{G}(\theta(y)), \Gamma(y) \rangle d\tau$  with  $w(\tau) = \frac{1}{2}(1 + \tau)^{-3/2}$ .

*Proof.* Each  $\theta_A(y) = \mathbb{E} g_A(y + Z)$ ,  $Z \sim N(0, I)$ , with  $g_A$  positively homogeneous of degree  $p_A \in \{1, 2\}$ , satisfies the heat equation  $\partial_s \theta_A = \frac{1}{2} \Delta \theta_A$  along  $N(y, sI)$  and the scaling  $\theta_A(y, s) = s^{p_A/2} \theta_A(y/\sqrt{s}, 1)$ , whence  $p_A \theta_A - y \cdot \nabla \theta_A - \Delta \theta_A = 0$ . By the chain rule  $\nabla e = \sum_A \partial_A \mathcal{G} \nabla \theta_A$  and  $\Delta e = \sum_A \partial_A \mathcal{G} \Delta \theta_A + \sum_{A,B} \partial_A \partial_B \mathcal{G} \Gamma_{AB}$ , so  $\delta = \mathcal{G} - \sum_A p_A \theta_A \partial_A \mathcal{G} - \sum_{AB} \partial_A \partial_B \mathcal{G} \Gamma_{AB}$ , and Euler's relation for the weighted homogeneity of  $\mathcal{G}$  kills the first two terms. The Gram entries follow from (3)–(4):  $\nabla_y \mu_a = g_a w_a$  and  $\nabla_y M_{ab} = T_{ab} w_a + T_{ba} w_b$ . The last statement is Corollary 1.3.  $\square$

Three remarks. (1) The theorem is exact and holds for every homogeneous chain (Gaussian closure or any cumulant chain run from the Gaussian surrogate of  $h_1$ ); nothing about the chain's internals enters except its curvature in the first-layer moments. (2) The Gram matrix has the *CP-leg* form of the cumulant chain: a gated covariance  $g \circ S \circ g$  (the arms), slice products  $T \circ S$  (the  $(2, 1)$  legs), and the four-term  $(M, M)$  block in which the pair  $(a, b)$  is linked to  $(c, d)$  through a single covariance entry. The same construction at a hidden layer  $l$  (tilt the hidden mean  $m_l$  with the chain's  $C_l$  in place of  $I$ ) gives the error of the chain started from the surrogate at layer  $l$ ,  $\text{Err}_l$ , and the differences  $\text{Err}_{l-1} - \text{Err}_l$  are the injected errors of Section 7. (3) The contraction  $\langle \nabla^2 \mathcal{G}, \Gamma \rangle$  can be evaluated inside the chain by propagating the second-order response of the closure to a Gaussian shift of the mean, i.e. the covariances  $\mathbb{E}[\delta m \delta m^\top]$ ,  $\mathbb{E}[\delta m_a \delta S_{ab}]$ ,  $\mathbb{E}[\delta m_a \delta S_{bb}]$ ,  $\mathbb{E}[\delta S_{aa} \delta S_{bb}]$ ,  $\mathbb{E}[\delta S_{ab} \delta S_{aa}]$  of the perturbation; every one of them closes at  $O(n^3)$  per layer through (3)–(6) except  $\mathbb{E}[\delta S_{ab}^2]$ , the variance of the transported cross-covariance, whose exact propagation is  $O(n^4)$ . This is the same obstruction that forces a cumulant chain to carry slices of  $\kappa_3$  rather than the tensor, met here from the other side: the second variation of a chain has the complexity of the chain's own closure error.

### 2.3 The organisers' chain in this language

The reference implementation of [2] (`alignment-research-center/mlp.cumulant.propagation`, which we ported to numpy and matched to  $10^{-15}$  on a small network, `whest/kprop3.py`) is the first-order theory of this section written as a term table. Its *Wick coefficients*  $\mathbb{E}[\partial^k \text{relu}(Z)^p]$  are the index shifts of Lemma 2.1: for  $p = 1$  they are  $\sigma^{1-k} d_k(\alpha)$ , and for  $p = 2, 3, 4$  the Hermite coefficients of  $\text{relu}^p$  (`relu2_hermite`, `relu3_hermite`). Its nonlinear step at  $k_{\max} = 3$  sums 36 diagrams (`notes/stage9/terms_k3_simple.tex`): for each output slice (powers  $p_i$  of the distinct indices) a product of cumulant blocks of  $z$  against the Wick coefficients  $\mathbb{E}[\partial^{k_i} \text{relu}^{p_i}]$  with  $k_i$  the multiplicity of index  $i$  over the blocks, cut off by the power counting  $\sum_{\text{blocks}} (|v| - 1 - [v \text{ even}]) \leq k_{\max} - 1$ . Read through Theorem 2.2, every term is a second- or third-order tilt response: the gate  $\Phi_i \Phi_j$  on  $C_{ij}$  is the gradient block  $\Gamma_{\mu\mu}$ , the birth  $w_j$  with two arms  $C_{ij} C_{jk}$  is the star of the cumulant chain and the diagonal of the Hessian block  $\Gamma_{MM}$ , and the coefficient  $\frac{1}{2} \Phi_i w_j$  of  $\kappa_3(z_i, z_i, z_j)$  in

$\text{Cov}(h_i, h_j)$  is (9). Two features of the reference matter for the system. First, in simple mode the slice gates are taken at  $\rho = 0$  and the pair moment is cut at  $O(\rho^2)$ : the  $\rho$ -dependence of a gate times a cumulant is a cost-3 diagram, dropped by the power counting; our chain’s full Hermite series of the Gaussian part is harmless but not what distinguishes the two. Second, the fourth cumulant is kept only as its radial (isotropic) part,  $\kappa_4(h) = c \text{Sym}(I \otimes I)$  with one scalar  $c = \frac{3}{n(n+2)} (\sum_{a \neq b} \kappa_4(h_a, h_a, h_b, h_b) + \sum_a \kappa_4(h_a))$  per layer, transported as  $c \text{Sym}(S \otimes S)$  with  $S = WW^\top$ : this is the scale mixture of Lemma 4.1, the central variable, and nothing else of  $\kappa_4$ . The reference thus confirms the split of Section 4 from the other side: the one piece of  $\kappa_4$  worth carrying is the collective scale.

### 3 Capacity of input tilts

The stage-8 programme asks first which localization sees the error. Here we settle it for the input side, with two measurements on a small network ( $n = 256$ ,  $L = 4$ , Monte Carlo truth from  $10^7$  samples, noise  $2 \cdot 10^{-4}$ ) and one on the official network 0 ( $n = 1024$ ,  $L = 16$ ).

#### 3.1 The Laplacian is the error; its directional parts are not

Table 1 gives, for the Gaussian closure and for the chain of this note (k3v3), the per-neuron correlation between the error  $e(0) - u$  and the Euler–Stein defect  $\delta(0) = e(0) - \Delta e(0)$  of Corollary 1.3, the least-squares coefficient  $a$  in  $e(0) - u \approx a \delta(0)$ , and the mean squared error after the correction  $e(0) - a\delta(0)$ . The Laplacian was computed exactly by  $2n + 1$  chain evaluations. Two facts. First,

| chain            | MSE                  | $ \delta _{\text{rms}}$ | $\text{corr}(e - u, \delta)$ | $a$   | fraction explained | MSE after $e - a\delta$ |
|------------------|----------------------|-------------------------|------------------------------|-------|--------------------|-------------------------|
| Gaussian closure | $1.16 \cdot 10^{-5}$ | $9.7 \cdot 10^{-3}$     | 0.967                        | 0.339 | 0.936              | $7.5 \cdot 10^{-7}$     |
| k3v3             | $1.09 \cdot 10^{-6}$ | $6.1 \cdot 10^{-3}$     | 0.835                        | 0.143 | 0.690              | $3.4 \cdot 10^{-7}$     |

Table 1: Euler–Stein defect against the true error,  $n = 256$ ,  $L = 4$  (`scripts/defect.small.py`).

$a < 1$ : Corollary 1.3 integrates  $\delta(y)$  against  $y \sim N(0, \tau I)$  with the weight  $\frac{1}{2}(1 + \tau)^{-3/2}$ , and  $\delta(y)$  decays as  $|y|$  grows (a shifted input makes the network more nearly deterministic relative to its fluctuations), so  $\delta(0)$  overstates the integral;  $a$  is the ratio of the weighted mean of  $\delta$  to its value at the origin, a property of the architecture class rather than of the network, and is fitted once. Second, the k3v3 chain, which carries the third cumulants, has a smaller  $a$  and a smaller explained fraction: its defect is of higher order and less self-similar in  $y$ .

The directional second differences tell the opposite story. For a unit vector  $v$  let  $\partial_v^2 e$  be the second difference of  $e$  along  $v$  at 0 and  $x_v = \partial_v^2 e - \Delta e/n$  its excess over the isotropic share. On the same network the singular values of the matrix  $(x_{e_i})_{i \leq n}$  decay slowly (top share 5%, top four 17%), the leading direction is orthogonal to the coherent direction  $W_1^\top \mathbf{1}$  ( $|\langle u_1, v_{\text{coh}} \rangle| < 10^{-3}$ ), and the correlation between  $x_v$  and the error is 0.18, 0.12, 0.08 (coherent, random, leading) for the Gaussian closure and 0.26,  $-0.05$ ,  $-0.15$  for k3v3. No input direction carries the error.

#### 3.2 The defect is a small residual of an $O(1)$ identity

At  $n = 1024$ ,  $L = 16$  (official network 0) the random-probe estimate of the Laplacian,  $\Delta e \approx \frac{1}{S} \sum_s \partial_{\xi_s}^2 e$  with  $\xi_s \sim N(0, I)$  (Hutchinson), fails outright: with  $S = 16$  probes per block the fitted coefficient is  $a \approx 0.01$  and the correlation with the error changes sign from block to block

(+0.43, -0.09, -0.44, +0.14). The reason is quantitative. Euler–Stein says  $\Delta u = u$  for the exact answer, and the chain nearly obeys it:  $\Delta e(0) = e(0) - \delta(0)$  with  $|\delta| \approx 10^{-2}$  against  $|e| \approx 0.4$ . The quantity we need,  $\delta$ , is a 2% residual of an  $O(1)$  identity, and the error is a tenth of that residual, so the Laplacian must be computed to a relative precision of  $10^{-4}$ . For a unit direction  $v$  one measures  $|\partial_v^2 e|_{\text{rms}} \approx 4 \cdot 10^{-3}$  at  $n = 256$ ; since  $\mathbb{E}_v(\partial_v^2 e)^2 - (\text{tr } H/n)^2 = 2\|H\|_F^2/n^2$  for  $v$  uniform on the sphere, the Hessian  $H = \nabla^2 e$  of one output neuron has  $\|H\|_F \approx n|\partial_v^2 e|/\sqrt{2} \approx 0.8$ , comparable with  $\text{tr } H \approx 0.4$ . The variance of one Hutchinson probe is  $2\|H\|_F^2$ , so the relative noise of an  $S$ -probe estimate of  $\text{tr } H$  is  $\sqrt{2/S} \|H\|_F / \text{tr } H \approx 2.8/\sqrt{S}$ , and the required  $10^{-4}$  needs  $S \approx 10^9$  probes. Only the exact coordinate sum,  $2n + 1$  chain evaluations, resolves  $\delta$ .

**Proposition 3.1** (Capacity of input tilts). *Let  $E$  be a chain exact on point masses and homogeneous as in Corollary 1.3, and let  $H$  be the Hessian of  $e$  at 0 for one output neuron. (i) The error is the trace part:  $u - e(0) = -\int w(\tau) \mathbb{E}_y \delta(y) d\tau$  with  $\delta(0) = e(0) - \text{tr } H$ . (ii) A localization that drives  $k$  orthonormal directions  $v_1, \dots, v_k$  (directional tilts, or pinning of  $k$  coordinates) has drift  $-\sum_{s \leq k} (\frac{1}{2} \partial_{v_s}^2 E - \partial_{\Sigma v_s} E)$ . Over a random choice of the  $k$  directions its expectation is  $(k/n)$  times the isotropic drift, whose error-carrying part is  $(k/n) \delta$ , while its fluctuation over the choice is of order  $\sqrt{k} \|A\|_F / n$  with  $A = \frac{1}{2} H - \partial_{\Sigma} E$ . The signal-to-noise ratio of a  $k$ -direction scheme is therefore  $\sqrt{k} |\delta| / \|A\|_F \approx 10^{-2}$  as measured, a  $k$ -direction scheme resolves the error only for  $k \gtrsim 10^4 \gg n$ , and the cheapest exact scheme on the input side is the coordinate sum,  $2n + 1$  chain evaluations, i.e.  $2n + 1$  pair chains at  $n = 1024$ , far outside any budget. Input localization is useless for estimation; it is a diagnostic.*

*Proof.* (i) is Corollary 1.3. (ii): the directional drift of Corollary 1.4 is a quadratic form in  $v$  with matrix  $\frac{1}{2} H - \partial_{\Sigma} E$ , whose average over the Haar measure on  $k$ -frames is  $k/n$  of its trace, and whose variance over the frame is  $\sum_s \text{Var}(v_s^\top A v_s) \approx 2k \|A\|_F^2 / n^2$  for  $A$  the matrix of the form. (iii) follows by comparing the two.  $\square$

This is the negative half of the stage-8 dictionary: the input is the *commutative* side, its tilts are the modular tilts of the Gaussian, and the chain’s defect is invisible to any finite family of them because it is a trace. The positive half is that the trace can be computed *inside* the chain, as the second-order response of the closure to a Gaussian shift of the mean, which is the subject of Sections 2 and 4: there the directions are neurons, the tilt responses are the births at the kinks, and the sum over directions is a sum over neurons that the chain already performs.

### 3.3 Why the merge coefficient is one half

On the official networks the fitted coefficient in  $u \approx e(0) - a \delta(0)$  is  $a = 0.535$  (network 0) and 0.500 (network 1), and the parameter-free midpoint  $(e + \Delta e)/2$  is as good as the fit (Section 7). Corollary 1.3 says  $a = \int_0^\infty \frac{1}{2} (1 + \tau)^{-3/2} g(\tau) d\tau$  with  $g(\tau) = \langle \mathbb{E}_{y \sim N(0, \tau I)} \delta(y), \delta(0) \rangle / |\delta(0)|^2$ , so  $a = \frac{1}{2}$  exactly when  $g(\tau) = (1 + \tau)^{-1/2}$ . That is the decay of the kink density: a mean shift  $y \sim N(0, \tau I)$  gives the first layer pre-activation means  $m_1 = W_1 y$  with variance  $2\tau$  against a variance 2 of the fluctuations, so  $\alpha_1 \sim N(0, \tau)$  and the birth strength averages to  $\mathbb{E} \varphi(\alpha) = (2\pi(1 + \tau))^{-1/2}$ , i.e.  $(1 + \tau)^{-1/2}$  times its value at the origin; by homogeneity the same holds at every layer, and a chain whose defect is first order in the births inherits the factor. The measurement on the small network (`scripts/defect_profile.py`,  $n = 256$ ,  $L = 4$ , Gaussian closure, three samples per  $\tau$ ) gives  $|\mathbb{E}_y \delta(y)| / |\delta(0)| = 0.96, 0.73, 0.46, 0.28$  at  $\tau = 0.25, 1, 4, 16$  against  $(1 + \tau)^{-1/2} = 0.89, 0.71, 0.45, 0.24$ , while the projection  $g(\tau) = 0.91, 0.56, 0.15, 0.02$  decays faster because at this small size the direction of  $\delta(y)$  rotates away from  $\delta(0)$ ; integrating the measured  $g$  against the weight gives  $a \approx 0.34$ ,

the value fitted in Table 1. At  $n = 1024$ ,  $L = 16$  the direction is stable,  $a$  is one half, and the *Euler–Stein midpoint* needs no constant: the chain’s error is the deficit of its births, and the births decay along the path exactly as the heat kernel’s weight requires. A chain whose defect is of order  $p$  in the births decays like  $(1 + \tau)^{-p/2}$  and has  $a = 1/(p + 1)$ : one half for the Gaussian closure ( $p = 1$ ), one third for a first-order chain ( $p = 2$ ; measured 0.255 for the organisers’ chain in Section 7). The merge coefficient reads off the order of a chain’s defect.

## 4 The collective mode is the central variable

The stage-8 principle splits a localization into its trivial (central) and Cartan parts. For the MLP the central variable is the collective activity of a layer: the direction  $s = \mathbb{E}[z^2]$  in pre-activation space, which is also the top eigenvector of the covariance once depth has built it up. This section proves two small exact statements about it and reports what the chain does to it.

### 4.1 A scale mixture is repaired exactly at first order by the (2, 2) channel

**Lemma 4.1** (Scale mixture). *Let  $z \mid \varepsilon \sim N(m, (1 + \varepsilon)^2 S)$  with  $\mathbb{E}\varepsilon = 0$ ,  $\mathbb{E}\varepsilon^2 = v$  small. Then the cumulants of  $z$  are those of  $N(m, (1 + v)S)$  at second order and  $\kappa_4(z_a, z_b, z_c, z_d) = 4v(S_{ab}S_{cd} + S_{ac}S_{bd} + S_{ad}S_{bc}) + O(v^2)$ , a pure (2, 2)-type tensor. The Gaussian closure applied to the true covariance  $(1 + v)S$  gives  $\mathbb{E}\text{relu } z_a$  with error  $-\frac{v}{2}\sigma_a(\alpha_a^2 - 1)\varphi(\alpha_a) + O(v^2)$ , and the  $\kappa_4$  term of the mean readout (10) with  $\kappa_4(z_a) = 12v\sigma_a^4$  is  $+\frac{v}{2}\sigma_a(\alpha_a^2 - 1)\varphi(\alpha_a)$ : the two cancel.*

*Proof.*  $\mathbb{E}[(1 + \varepsilon)^4] = 1 + 6v + O(v^2)$  and  $\mathbb{E}[(1 + \varepsilon)^2]^2 = 1 + 2v + v^2$  give the fourth cumulant. Put  $f(\varepsilon) = (1 + \varepsilon)\sigma d_0(\alpha/(1 + \varepsilon))$ ; the true mean is  $\mathbb{E}f(\varepsilon) = f(0) + \frac{v}{2}f''(0)$  and the closure’s is  $f(\varepsilon^*)$  with  $(1 + \varepsilon^*)^2 = 1 + v$ , i.e.  $f(0) + \frac{v}{2}f'(0)$ . From Lemma 2.1,  $f'(0) = \sigma[d_0 - \alpha d_1] = \sigma\varphi$  and  $f''(0) = \sigma\alpha^2\varphi$ .  $\square$

So the (2, 2) channel of the chain (k22:  $\kappa_4(h_a, h_a, h_b, h_b)$  born exactly from the Hermite series of  $\text{Cov}((h_a - \mu_a)^2, (h_b - \mu_b)^2)$ , transported by  $(W \circ W)K(W \circ W)^\top$  with the gate  $tt^\top$  and the diagonal class removed once) is the first-order correction for a fluctuating collective scale. Its gain projection  $4v = s^\top K_{22}s/(s^\top s)^2$  measures  $v$  directly.

### 4.2 The sensitivity law

**Proposition 4.2** (Sensitivity). *A relative error  $\varepsilon_a$  in the pre-activation variance of neuron  $a$  changes its post-activation mean by  $\frac{1}{2}\varepsilon_a\sigma_a\varphi(\alpha_a)$ , and an error  $\delta\lambda$  in the top eigenvalue of the covariance of layer  $l$  (eigenvector  $u$ ) changes the pre-activation variance of every neuron  $b$  of layer  $l + 1$  by  $\delta\lambda(w_b \cdot u)^2 \approx 2\delta\lambda/n$ . On the official networks ( $\sigma \approx 2.8$ ,  $\varphi(\alpha) \approx 0.3$ ,  $\text{var} \approx 8$ ,  $\lambda_1 \approx 12$  at depth) a uniform relative variance error of  $10^{-3}$  moves every mean by  $4 \cdot 10^{-4}$ , seven times the rms error of the chain, while a 2% error of the top eigenvalue moves the next layer’s variances by  $5 \cdot 10^{-5}$  relative and the means by  $2 \cdot 10^{-5}$  per layer.*

*Proof.*  $\partial_{\sigma^2}[\sigma d_0(m/\sigma)] = \frac{1}{2\sigma}(d_0 - \alpha d_1) = \varphi/(2\sigma)$ , multiplied by  $\varepsilon\sigma^2$ ; and  $w_b^\top(\delta\lambda uu^\top)w_b$  with  $w_b \cdot u \sim N(0, 2/n)$  for He rows.  $\square$

The bulk of the covariance may therefore be wrong by percents without consequence (it enters the next layer’s variances through  $\sum_c S_{bc}^2$ -type sums that average out), while the diagonal must be right to  $10^{-4}$  and the collective mode to  $10^{-3}$ : that is what makes the central variable the one that matters.

### 4.3 Measurements on the official network 0

From the Monte Carlo cache ( $2^{20}$  inputs; `scripts/diag-eig-v3.py`, `diag-onestep.py`):

- The collective mode emerges with depth: the top four eigenvalues of the pre-activation covariance are 8.08, 7.89, 7.77, 7.71 at layer 0 and 16.3, 5.36, 4.88, 4.62 at layer 15.
- The chain’s trace is exact to  $10^{-3}$  at every layer and its diagonal to  $10^{-4}$ ; the bulk (the complement of the top four modes, off-diagonal part) is off by 2–4%; the top eigenvalue is correct to  $2 \cdot 10^{-3}$  through layer 5 and then 1.0, 1.2, 1.5, 1.8, 1.9, 1.8, 1.7, 1.8, 1.5, 1.6% *low* at layers 6–15, with the variance along  $s$  equally low. By Proposition 4.2 this is a  $2 \cdot 10^{-5}$  per layer effect on the means, a tenth of the measured error: the top mode is the *symptom* of the deficit, not its carrier.
- One-step closure from Monte Carlo inputs at layers 11, 13, 14: the Gaussian closure loses 3.2, 1.5, 1.1% of the top mode in one layer; adding the Monte Carlo (2, 1) slices (the  $\kappa_3$  covariance correction (9)) leaves  $-0.13, -0.09, -0.09\%$ ; adding the (2, 2) slices changes that by less than  $3 \cdot 10^{-4}$ ; adding the (3, 1) slices makes it worse ( $-0.19, -0.15, -0.16\%$ ). The  $\kappa_3$  slice is the whole one-step story of the collective mode, and the sign of the (3, 1) term says its first-order transport is not the right object to add.
- The chain’s own transported slices at those layers have correlation 0.965 with Monte Carlo and slope 0.92, 0.90, 0.91; its (2, 2) slices have correlation 0.22–0.25 and slope 0.53–0.68, and its gain projection  $4v$  is 3.6, 1.9,  $1.5 \cdot 10^{-5}$  against 6.6, 4.6,  $4.0 \cdot 10^{-5}$ : the scale-mixture variance is  $v \approx 10^{-5}$ , so by Lemma 4.1 the (2, 2) channel is worth  $\sim 10^{-5}$  per neuron and layer, which is why the chain’s (2, 2) errors are harmless.

The error accounting that follows is then simple. A 10% deficit of the transported slice leaves 10% of the 1–3% one-step loss of the collective mode in place, 0.1–0.3% per layer, which accumulates to the 1.5–1.9% measured; the same deficit leaves 10% of the  $\kappa_3$  corrections to the diagonal and the means in place at every layer, and the per-layer injected mean errors of  $3 \cdot 10^{-4}$  (Section 7) are of that size: 10% of  $\kappa_3(z_a)\alpha\varphi/(6\sigma^2)$  at  $\kappa_3 \approx 3$ , which is  $3 \cdot 10^{-4}$ . The chain’s error is the deficit of its (2, 1) transport, in the diagonal readout and in the collective mode alike.

## 5 Exact first-order transport through ReLU

The chain of this note (`whest/k3chain3.py`) transports the third cumulant in the CP-leg form of [2]: each kink neuron  $m$  of layer  $l$  is a source with centre leg  $P = W \cdot e_m$ , arm  $A = W \cdot a_m$ ,  $a_m = \text{diag}(g) C e_m$ , strength  $w_m = \varphi(\alpha_m)/\sigma_m$ , and the two further legs  $B$  (the Gram–Charlier arm fed by the incoming (2, 1) slices) and  $R$  (the residual that makes the slice and the diagonal exact at birth). The readouts in pre-activation space are the diagonal  $\kappa_3(z_a) = D3_a$  and the slices  $D21_{ab} = \kappa_3(z_a, z_a, z_b)$ . What is new relative to [2] is the *transport* of a slice through a ReLU layer to first order in the cumulants, in closed form, which we state and then measure.

**Theorem 5.1** (First-order transport of the (2, 1) slice). *Let  $z \sim$  a law with Gaussian part  $N(m, S)$  and small third and fourth cumulants,  $h = \text{relu}(z)$ , and write  $g = \Phi(\alpha)$ ,  $w = \varphi(\alpha)/\sigma$ ,  $\mu_a = \sigma_a d_0(\alpha_a)$ ,  $t_a = g_a - \mu_a w_a$ ,  $\epsilon_a = 2\mu_a(1 - g_a)$ ,  $c_a = g_a(1 - g_a) - \mu_a w_a$ . To first order in the*

cumulants of  $z$  and to the orders in  $\rho$  kept by the chain,

$$\kappa_3(h_a, h_a, h_c) = \left[ g_a^2 + c_a \right] g_c \kappa_3(z_a, z_a, z_c) + \frac{1}{2} \epsilon_a w_c \kappa_3(z_a, z_c, z_c) + \frac{1}{2} t_a w_c \kappa_4(z_a, z_a, z_c, z_c) + (\text{born}), \quad (8)$$

$$\text{Cov}(h_a, h_b) = \text{Cov}^{\text{Gauss}}(h_a, h_b) + \frac{1}{2} (w_a g_b \kappa_3(z_a, z_a, z_b) + g_a w_b \kappa_3(z_a, z_b, z_b)) + \frac{1}{4} w_a w_b \kappa_4(z_a, z_a, z_b, z_b), \quad (9)$$

$$\mathbb{E}[h_a] = \sigma_a d_0(\alpha_a) - \frac{\alpha_a \varphi(\alpha_a)}{6\sigma_a^2} \kappa_3(z_a) + \frac{(\alpha_a^2 - 1) \varphi(\alpha_a)}{24\sigma_a^3} \kappa_4(z_a), \quad (10)$$

where “born” is the exact  $\kappa_3$  of `relu` of the Gaussian part (all Hermite orders of the slice, `relu2_hermite`, and the exact diagonal `k3_exact`), and the gate of the transported slice,  $g_a^2 + c_a = g_a - \mu_a w_a$ , is the exact coefficient  $\mathbb{E}[\theta_a \partial_a (\text{relu } z_a - \mu_a)^2]/2$  rather than the Gaussian  $g_a^2$ .

*Proof sketch.* Expand the law of  $z$  as a Gram–Charlier perturbation of its Gaussian part,  $\nu = \nu_G(1 + \frac{1}{6} \kappa_3 : \text{He}_3 + \frac{1}{24} \kappa_4 : \text{He}_4)$ , and compute  $\mathbb{E}[(h_a - \mu_a)^2 (h_c - \mu_c)]$  by integrating the Hermite polynomials against the ReLU moments: Stein’s lemma turns each  $\text{He}_3$  against a product of ReLUs into third derivatives, which are sums of gate products  $\theta_a \theta_a \theta_c$  (the first term), of  $\theta_a \delta_c \text{relu}_c$ -type terms, i.e. a birth  $w_c$  at  $c$  against the transposed slice (the second term), and the  $\text{He}_4$  term gives the  $(2, 2)$  feed with the single birth  $w_c$  (third term). The coefficients  $t_a, \epsilon_a, c_a$  are the exact one-dimensional integrals  $\mathbb{E}[\theta_a (\text{relu } z_a - \mu_a) \cdot]$  listed in the code. The covariance and mean formulas are the same computation one order lower.  $\square$

The three transported terms are what the chain calls the gated slice (`gate_res`), the transposed-slice feed (`feedT`) and the  $\kappa_4 \rightarrow \kappa_3$  feed (`feedK22`). Their effect on the official network 0 is the ladder of Section 7: the transported slices rise from 75% to 90–92% of their Monte Carlo values at depth and the final MSE falls from  $8.5 \cdot 10^{-7}$  to  $5.7 \cdot 10^{-7}$ .

**What remains open in the transport.** The first term of (8) transports the slice with the gate of the pair  $(a, c)$  taken at  $\rho_{ac} = 0$  plus the chain’s  $\rho$ -expansion; the exact gate  $\mathbb{E}[\theta_a \theta_a \theta_c(\cdot)]$  at the true correlation differs at order  $\rho_{ac}$ , which is small per entry but coherent along the collective mode. Section 9 lists the  $\rho$ -exact gate as the first open item, since the 8–10% deficit of the transported slice at depth is precisely of this size.

## 6 What the chain costs: the pair floor

The score multiplies the error by  $\max(0.1, C/B)$  with  $C$  the metered count of the grader’s own numpy wrapper (`flopscope 0.12.1`) and  $B = 2^{41}$ . We read the wrapper’s rules off its source and measured the bills of the chain’s building blocks at  $n = 1024$  (`notes/compute/FLOPSCOPE.md`); the facts that decide a design are these.

**Rules.** Every call bills  $\lfloor \text{textbook count} \times \text{dtype rate} \times \text{weight} \rfloor$ . The count is the textbook one, not what BLAS does: a `matmul(m, k)(k, n)` is  $2mkn - mn$  whatever algorithm the library uses, so a Strassen level written out of seven smaller `matmul` calls is billed at 7/8. The dtype rate is 1 for float32 and 2 for float64. The weight is 0 for views, transposes, `zeros`; 1 for arithmetic, reductions, contractions, copies, `sqrt`, `stats.norm.cdf/pdf` (whose counts are 48 and 27 per element, computed in float64), `polyval` ( $2d$  per element) and all of `linalg`; 4 for `where`, gathers and sorts; 16 for `exp`, `log` and every random sampler. A contraction whose output the

engine can prove symmetric is billed on the unique entries: the sandwich  $WCW^\top$  written as one `einsum('ij,jk,lk->il', W, C_sym, W)` with the same array  $W$  in both outer slots and  $C$  tagged `as_symmetric` bills  $3n^3$  in float32 ( $6n^3$  in float64) against  $4n^3$  ( $8n^3$ ) for two matmuls; the saving is lost if the two products are separate calls. `eigh` is  $9n^3$ , `cholesky`  $n^3/3$ , `inv`  $2n^3$ , `solve`  $\frac{2}{3}n^3 + 2n^2r$ .

**Budget.** The floor  $0.1B = 2.2 \cdot 10^{11}$  equals  $205n^3$  over the 16 layers, i.e.  $12.8n^3$  per layer: six float32  $(n, n, n)$  matmuls, or four float32 symmetric sandwiches, or three float64 matmuls. Nothing is gained below the floor, so a chain should spend exactly it. The Gaussian pair chain, one sandwich per layer, costs  $0.024B$  in float32 (one quarter of the floor) and  $0.047B$  in float64. Elementwise work on  $n \times n$  arrays is negligible unless repeated dozens of times per layer: the Hermite series of the pair moment to order  $K$  is  $K$  Hadamard products,  $2Kn^2$ , i.e.  $0.002n^3$  per term.

**Proposition 6.1** (Pair floor). *Any estimator that forms the pre-activation covariance of every layer from that of the previous one pays at least one sandwich per layer:  $3n^3$  in float32 with the symmetric engine,  $L \cdot 3n^3 = 0.024B$  at  $n = 1024$ ,  $L = 16$ . Every further  $n \times n$  object transported through  $W$  (a  $\kappa_3$  slice, a  $(2, 2)$  slice, a gate-correlation matrix) costs another  $2n^3$  to  $3n^3$  per layer, i.e.  $0.016B$  to  $0.024B$  over the network. Under the floor a chain may therefore carry the pair plus about four such objects per layer; the leaders' bills ( $0.105B$ ,  $0.144B$ ) correspond to five to seven float32 sandwich-equivalents per layer, and the open-source third-cumulant chain at  $0.25B$  to a float64 implementation of the same count.*

**Naive counts mislead.** The count our chain returns (FMA = 2, no dtype rate, no symmetry) overstates the float32 symmetric bill by  $2.7\times$  and understates a float64 plain-matmul bill by  $2\times$ . The *pair floor* and the *per-object price* above are the only two numbers a design needs; every proposal in Section 9 is costed in units of “ $n \times n$  objects transported per layer”.

**Precision.** float32 arithmetic (relative  $10^{-7}$  per operation) propagates the covariance through sixteen layers of 1024-term sums to a relative  $10^{-6}$ , far below the  $10^{-4}$  rms of a  $10^{-8}$ -MSE estimator; but the quantities that are small differences of  $O(1)$  numbers (the Euler–Stein defect  $e - \Delta e$ , differences of nearly equal slices) must be formed in float64 or arranged to avoid cancellation.

## 6.1 The memory of the third cumulant and the transport cocycle

The organisers' chain keeps the all-distinct part of  $\kappa_3$  in factored form  $\text{Sym} \sum_r A_{ir} B_{jr} C_{kr}$  and adds  $2n$  columns to each leg at every layer (the star block of the births and the residual block that makes the  $(2, 1)$  and  $(3)$  slices exact), so the rank at layer  $l$  is  $2ln$  and the transport and readout cost  $18ln^3$  per layer,  $1.2B$  in float32 over the network: 93% of the reference's own count. The theory says where to cut. A block born at layer  $l'$  is, at layer  $l$ , the image of its birth legs under the gated transport cocycle

$$T_{l \leftarrow l'} = W_l \text{diag}(g_{l-1}) W_{l-1} \text{diag}(g_{l-2}) \cdots W_{l'+1} \text{diag}(g_{l'}),$$

a product of  $l - l'$  independent gated random matrices. Such products contract onto their leading Oseledets directions, and the leading directions are the collective mode of Section 4: in the stage-8 language the old sources live on the leaf space, and the shared basis of their transport is the transverse measure's support. On network 0 (`scripts/diag_cocycle.py`) the effective rank

$\|T\|_F^2/s_1^2$  of  $T_{l \leftarrow l-a}$  is 164, 92, 65, 47, 29, 23 at ages  $a = 1, 2, 3, 4, 6, 8$  (independent of  $l$  to a few percent), and a random block transported for  $a$  ages has 51, 70, 81, 87, 94, 96% of its energy in the top 128 left singular directions and 77, 92, 95, 98.5, 99.7, 99.8% in the top 256. Sources of age  $\geq 2$  can therefore be merged into a *Tucker tier*  $\text{Sym}((Q \otimes Q \otimes Q)S)$  with one orthonormal basis  $Q \in \mathbb{R}^{n \times k}$  shared by the three legs and a symmetric core  $S \in \mathbb{R}^{k \times k \times k}$  (`whest/kprop3c.py`): the basis is transported as  $W \text{diag}(g)Q$  and re-orthonormalised, the core rotated by the  $k \times k$  factor, and an aged block is folded in by refitting the basis to the dominant directions of  $[Q \text{diag}(\nu), A, B, C]$  and projecting. Its cost per layer is  $2nk^3 + 4n^2k$  for the two slice readouts and  $2nk^3$  for a merge, i.e.  $19n^3$  at  $k = 128$  with no young tier, against  $18ln^3$  dense; at  $k = 256$  the cubic core ( $2nk^3 = 32n^3$  per readout) is already too expensive, which is the statement that a generic order-three tensor on a  $k$ -dimensional leaf space has  $k^3/6$  degrees of freedom and cannot be read for less. The young tier (ages  $< w$ ) costs  $16n^3$  per source-age. Table 2 gives the resulting bills under the grader’s rules; the measured accuracy of each configuration is in Section 7.

| $(w, k)$ | $n^3/\text{layer}$ | $C/B$ (float32) | $(w, k)$        | $n^3/\text{layer}$ | $C/B$ (float32)            |
|----------|--------------------|-----------------|-----------------|--------------------|----------------------------|
| (0, 128) | 19                 | 0.148           | (2, 128)        | 51                 | 0.398 (0.32 with Strassen) |
| (1, 128) | 35                 | 0.273 (0.23)    | (3, 128)        | 67                 | 0.523 (0.41)               |
| (0, 256) | 92                 | 0.716           | dense reference | $18l$              | 1.195                      |

Table 2: Billed cost of the two-tier chain (pair chain, metric, young tier, Tucker tier, merge, elementwise terms).

## 6.2 Why the memory resists Frobenius compression

The measured accuracy of the two-tier chain (Section 7) is far worse than the energy fractions of the cocycle suggest, and the reason is structural. The slices are read through *Hadamard* products of the legs,  $(A \circ B)C^\top$ : for a residual block transported for  $a$  ages the  $(2, 1)$  slice is  $\frac{1}{3}[2((TR) \circ T)T^\top + (T \circ T)(TR)^\top]$  with  $T = T_{l \leftarrow l-a}$ . Projecting a leg on the top- $k$  subspace of  $T$  controls  $\|T - QQ^\top T\|_F$ , but the Hadamard square  $(QQ^\top T) \circ (QQ^\top T)$  is not the projection of  $T \circ T$ : the square of a rank- $k$  matrix has rank up to  $k(k+1)/2$  and lives in a different subspace. A leg truncation that loses a fraction  $\epsilon^2$  of the energy therefore loses a fraction  $\sim 2\epsilon$  of the slice, and by the sensitivity law of Proposition 4.2 the slices of the sources that matter must be right to  $10^{-3}$ : the leg energy must be captured to  $10^{-6}$ , which the cocycle spectrum ( $s_{256}/s_1 = 0.5, 0.3, 0.18, 0.17$  at ages 1–4) is nowhere near. Compression of the kind that works for a *linear* transport (keep the dominant directions of  $T$ ) does not work for a cubic readout, except for sources old enough that their whole contribution is small: at age  $\geq 4$  the sources are worth a factor 6 in MSE, so a 5% slice error on them costs  $\sim 10\%$ , which is the regime in which the open-source chain’s shared basis ( $k = 384$ , window 4) loses its reported 3.5%. The arithmetic of the pair floor then says that an exact-first-order chain cannot be brought near  $0.1B$  by compressing its memory: the young window alone, four ages at  $16\text{--}24n^3$  each, exceeds the whole budget. The birth blocks have identity legs, so each source-age needs only three transported matrices ( $T, TX_1, TR$ , with  $X_3 = X_1 \text{diag}(w)$  a column scaling of  $X_1$ ) and three readout products once the two terms with  $T^\top$  on the right are summed first:  $12n^3$  per source-age instead of 24, 8.3 with three Strassen levels. The exact first-order chain then costs  $12 \sum_l l n^3 \approx 0.7B$  ( $0.5B$  with Strassen) in float32, and a window of  $w$  ages  $12wn^3$  per layer plus whatever represents the rest; a  $k$ -dimensional CP tier costs about  $11n^3$  per old source-layer at  $k = 384$ , no less than the dense form, because the projection helps the transport and not the Hadamard readouts. What a competitive chain at  $0.1B$  must do instead

is keep *fewer* ages and recover the systematic part of what is lost by calibration on the public networks; Section 7 measures how much of the loss is systematic (a scale of the old sources’ slices) and how much is shape.

## 7 Measurements on the official networks

All numbers are for the official network 0 ( $n = 1024$ ,  $L = 16$ , reference means from  $10^9$  Monte Carlo inputs) unless stated; network 1 is used as the held-out check where available.

### 7.1 The error ladder of the clean chain

| estimator                                                                                                | final-layer MSE                        | what is added                      |
|----------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------|
| Gaussian closure ( <code>gauss_chain</code> )                                                            | $4.06 \cdot 10^{-6}$                   | —                                  |
| $\kappa_3$ CP legs, stars only, no hub                                                                   | $3.76 \cdot 10^{-6}$                   | births $w$ , arms d                |
| + hub: $D21 \rightarrow$ covariance (9), GC arm                                                          | $8.37 \cdot 10^{-7}$                   | transported                        |
| + exact birth slice and diagonal (residual leg $R$ ), exact gates                                        | $8.52 \cdot 10^{-7}$                   | Theorem 5.1, f                     |
| + transposed-slice feed, $\kappa_4 \rightarrow \kappa_3$ feed, (2, 2) channel, exact diagonal $\kappa_3$ | $5.74 \cdot 10^{-7}$                   | Theorem 5.1, second                |
| Gaussian closure + Euler–Stein defect, $e - a\delta(0)$ , $a = 0.535$                                    | $4.35 \cdot 10^{-7}$                   | Corollary 1.3, exact $\delta$      |
| organisers’ chain ( <code>kprop3</code> ) + Euler–Stein defect, $e - a\delta(0)$ , $a = 0.255$           | <b><math>6.64 \cdot 10^{-9}</math></b> | same: corr 0.90, 81                |
| Gaussian closure + midpoint $(e + \Delta e)/2$ (no fitted constant)                                      | $4.50 \cdot 10^{-7}$                   | same                               |
| organisers’ chain, $k_{\max} = 3$ simple, numpy port ( <code>kprop3</code> ), radial $\kappa_4$ off      | $5.63 \cdot 10^{-7}$                   | exact first-order $\kappa_3$ trans |
| same, radial $\kappa_4$ on (one scalar per layer)                                                        | $3.53 \cdot 10^{-8}$                   | the central variable               |
| open-source third-cumulant chain (504aldo), float32, 33 fitted scalars, $0.25B$                          | $2.13 \cdot 10^{-8}$                   |                                    |
| exact first-order $K = 3$ reference of the organisers (no $\kappa_4$ , no fits)                          | $4.3 \cdot 10^{-8}$                    |                                    |
| leaders (raw)                                                                                            | $1.1\text{--}1.5 \cdot 10^{-8}$        |                                    |

The decisive line is the pair in the middle: the organisers’ exact first-order chain, ported to numpy and matched to the torch reference bit for bit, reaches  $5.6 \cdot 10^{-7}$  on network 0 when its fourth cumulant is dropped, the accuracy of our own chain, and  $3.5 \cdot 10^{-8}$  when the single radial scalar  $c = \frac{3}{n(n+2)}(\sum_{a \neq b} \kappa_4(h_a, h_a, h_b, h_b) + \sum_a \kappa_4(h_a))$  per layer is kept. One number per layer, the collective scale of Lemma 4.1, is worth a factor 16 in MSE; our chain’s transported (2, 2) slice, which carries  $n^2$  numbers and is harmful off the diagonal, was worth nothing. The reference’s own naive count is  $3.4 \cdot 10^{12}$  FLOPs ( $1.5B$  in float32), 93% of it in the transport and readout of the factored  $\kappa_3$  whose rank grows by  $2n$  per layer; the cost problem is the memory of the third cumulant, not the fourth.

On network 1 the same measurement gives correlation 0.951,  $a = 0.500$ , 93% explained and  $4.72 \cdot 10^{-6} \rightarrow 3.45 \cdot 10^{-7}$ ; the fitted coefficients cross over ( $a_0$  on network 1:  $3.68 \cdot 10^{-7}$ ;  $a_1$  on network 0:  $4.51 \cdot 10^{-7}$ ), and the parameter-free midpoint  $(e + \Delta e)/2$  gives 4.50 and  $3.45 \cdot 10^{-7}$ : the weight of Corollary 1.3 integrates to one half on these networks, so the *Euler–Stein midpoint* is an estimator with no fitted constant that is nine to fourteen times better than the chain it corrects.

For the exact first-order chain the same measurement (2049 runs of `kprop3` on Modal, 41 s each on four cores) gives  $|\delta(0)|_{\text{rms}} = 6.6 \cdot 10^{-4}$  against an error of  $1.9 \cdot 10^{-4}$ , correlation 0.902, 81% of the mean square explained,  $a = 0.255$ , and a corrected MSE of  $6.64 \cdot 10^{-9}$ : a factor 5.3 below the chain and a factor two below the leaders’ raw accuracy, from a correction that is estimator-agnostic and derived, not fitted, except for the one constant  $a$ . The midpoint  $a = \frac{1}{2}$  does not apply to this chain ( $3.3 \cdot 10^{-8}$ ): its defect is second order in the births, so it decays along the path like the square of the kink density,  $(1 + \tau)^{-1}$ , which integrates against the weight to  $\frac{1}{3}$ ,

and the measured 0.255 sits below that by the same direction rotation seen on the small network. The cost of this correction is  $2n + 1$  chain evaluations, which is why the symbolic second-order response of Theorem 2.2 is the open engineering problem of this note. The constant is a property of the chain, not of the network: on network 1 the same 2049 runs give correlation 0.914, 84% explained,  $a = 0.257$  and  $4.38 \cdot 10^{-8} \rightarrow 7.15 \cdot 10^{-9}$ , and the two constants cross over without loss ( $a_0$  on network 1:  $7.152 \cdot 10^{-9}$ ;  $a_1$  on network 0:  $6.646 \cdot 10^{-9}$ ). A single constant  $a \approx 0.256$ , fixed offline, corrects the exact first-order chain by a factor 5–6 on every network tried.

Two readings. First, the Euler–Stein defect of the *Gaussian* closure, with no cumulant machinery at all, reaches the accuracy of our third-cumulant chain: on network 0 the exact  $\delta(0)$  (coordinate Laplacian, 2049 chain evaluations on Modal) has correlation 0.931 with the error, explains 89% of its mean square, and the fitted  $a = 0.535$  is close to the parameter-free midpoint  $a = \frac{1}{2}$ . Theorem 1.2 is exact; the 11% not explained is the variation of  $\delta(y)$  over the Eldan path. Second, the organisers’ exact first-order  $K = 3$  reference is  $13\times$  better than our chain and carries no fitted constant, so the gap is in the first-order transport, not in calibration or in  $\kappa_4$ .

## 7.2 The memory of the third cumulant: compression and calibration

All runs on network 0 with the ported chain (`whest/kprop3c.py`); dense reference  $3.53 \cdot 10^{-8}$ .

| configuration                                                                        | final MSE                                                      |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------|
| dense (rank $2ln$ )                                                                  | $3.53 \cdot 10^{-8}$                                           |
| window 4, CP tier $k = 384$                                                          | $3.69 \cdot 10^{-8}$                                           |
| window 4, CP tier $k = 256$                                                          | $5.21 \cdot 10^{-8}$                                           |
| window 2, CP tier $k = 384$                                                          | $7.90 \cdot 10^{-8}$                                           |
| window 4, Tucker $k = 128$                                                           | $2.00 \cdot 10^{-7}$                                           |
| window 2, Tucker $k = 256$                                                           | $3.21 \cdot 10^{-7}$                                           |
| window 2, Tucker $k = 128$                                                           | $9.34 \cdot 10^{-7}$                                           |
| window 2, Tucker $k = 128$ , collective directions kept                              | $8.41 \cdot 10^{-7}$                                           |
| window 1, Tucker $k = 128$                                                           | $1.96 \cdot 10^{-6}$                                           |
| window 0, Tucker $k = 128$                                                           | $3.71 \cdot 10^{-6}$                                           |
|                                                                                      |                                                                |
| window 2, old sources dropped                                                        | $3.68 \cdot 10^{-6}$                                           |
| window 2, old sources: diagonal $\kappa_3$ only                                      | $1.03 \cdot 10^{-6}$                                           |
| window 2, old sources scaled by 0.9                                                  | $5.13 \cdot 10^{-6}$                                           |
| window 2, old sources’ slices replaced by their rank-1 / 4 / 16 / 64 truncation      | $2.24 \cdot 10^{-7} / 2.03 \cdot 10^{-7} / 1.56 \cdot 10^{-7}$ |
| window 1, rank 16 / 64; window 3, rank 4                                             | $3.05 \cdot 10^{-7} / 1.50 \cdot 10^{-7}$                      |
| window 4, old sources dropped                                                        | $7.45 \cdot 10^{-7}$                                           |
| no window: birth blocks sparsified to the 50% / 25% / 10% most important hub columns | $4.12 \cdot 10^{-7} / 1.53 \cdot 10^{-7}$                      |
|                                                                                      |                                                                |
| dense, radial scalar $\times 0.9 / \times 1.1 / \times 1.25$                         | $8.9 \cdot 10^{-8} / 2.7 \cdot 10^{-7}$                        |

The old content is not a scaled copy of the young content: regressing the  $(2, 1)$ -slice contribution of the sources of age  $\geq 2$  on that of age 1 (`scripts/diag_oldshape.py`) gives correlations 0.29–0.43 and a residual of 0.90–0.96 of its norm at every layer (diagonal  $\kappa_3$ : correlations 0.43–0.61), while the old content grows from 0.33 to 3.1 times the young content between layers 3 and 15. Memoryless regeneration of  $\kappa_3$  from the current layer with per-layer scalars would keep a tenth of it.

The old content splits into a coherent and an incoherent part. Replacing the old sources’  $(2, 1)$  slices by their best rank-one approximation recovers most of what dropping them loses ( $3.7 \cdot 10^{-6} \rightarrow 2.2 \cdot 10^{-7}$ , against  $3.5 \cdot 10^{-8}$  for the full memory), and the remainder converges slowly in rank (rank 64:  $7.8 \cdot 10^{-8}$ ). This is what the Hadamard structure predicts: for products of

gated random matrices the Hadamard squares split into a separable mean (rank one per term, computable from the variance profile of the cocycle and its action on a few vectors at  $O(n^2)$ ) and incoherent fluctuations, which decay with age but are still large at ages two to five.

The birth blocks are not sparsifiable in the hub index either. Keeping, in every star and residual block, only the  $qn$  columns of largest importance ( $w_{2,j}|a_j|^2|c_j|^2$  for the star, the column norm for the residual leg; `srcfrac` in `whest/kprop3c.py`) costs a factor 12 at  $q = \frac{1}{2}$  and 43 at  $q = \frac{1}{4}$ , and a least-squares rescaling of the kept columns changes nothing (the dropped and kept columns are orthogonal in the slice inner product, so the rescaling factor is one). The importance profile is flat: at He initialisation every hub neuron contributes a comparable, mutually orthogonal rank-one term, which is the regime in which subset selection cannot work (Section 8).

Three conclusions. (i) The old sources are essential and their effect is smooth in amplitude but not in shape: a 10% scale error on everything older than two layers costs a factor 1.5, dropping them costs 100, and a Frobenius truncation that keeps 70–87% of their energy ( $k = 128$ ) costs 26, exactly the Hadamard obstruction of Section 6. (ii) A window of four dense ages with a  $k = 384$  shared basis for the rest is lossless to 5% (two ages: a factor 2.2; four ages with  $k = 256$ : 1.5), reproducing the open-source chain’s regime, and costs about  $1B$  in float32 without the identity-leg and Strassen savings ( $\approx 0.45B$  with them); the floor is  $0.1B$ . (iii) The radial scalar is exactly right as projected and the chain is sharply sensitive to it ( $\pm 10\%$ :  $2.5\times$  and  $8\times$ ), so the central variable is not where calibration helps; float32 reproduces float64 to all digits shown.

### 7.3 Where the chain’s error is injected

With  $e_l = \text{out}_l - u_l$  and the injected part  $e_l - \Phi_l \cdot (W_l e_{l-1})$  (the error not explained by the linear propagation of the previous layer’s error through the gates), `scripts/injected.py` gives rms injected errors of 2.0, 1.1, 1.6, 2.1, 2.3, 2.5, 2.9, 2.7, 2.8, 2.9, 2.8, 2.7, 2.8, 2.9,  $3.0 \cdot 10^{-4}$  at layers 1–15 against a total rms error growing from  $2.0 \cdot 10^{-4}$  to  $7.6 \cdot 10^{-4}$ : the chain injects a nearly constant  $3 \cdot 10^{-4}$  per layer from layer 4 on, and the injected error has no coherent component along the mean direction (share  $< 10^{-2}$  beyond layer 2). This is the signature of a *per-neuron*, incoherent deficit in the transported slices, consistent with Section 4: the transported  $D21$  has correlation 0.965 with Monte Carlo, i.e. an incoherent relative error of  $\sqrt{1 - 0.965^2} \approx 26\%$  on top of the 8–10% coherent deficit, and by the empirical law of [2]’s reimplemention (extra MSE  $\approx 4 \cdot 10^{-6} \varepsilon^2$  for a relative rms slice error  $\varepsilon$ ) a slice error of 35% accounts for the whole  $5.7 \cdot 10^{-7}$ .

### 7.4 What the open-source chain does that ours does not

A reading of the open-source chain’s derivation notes against `k3chain3.py` (delegated, cross-checked on the formulas) locates the difference in exactly the first-order transport: (i) its slice update carries every elementwise diagram of first order in the cumulants, i.e. the  $\rho$ -dependence of the gates ( $D21_{ac} C_{ac}^k \times$  Wick coefficients) and the  $\kappa_3(z_a)$ -driven terms, where ours keeps the gates at  $\rho = 0$ ; each such term is  $O(\rho) \approx 3\%$  per layer and compounds over fifteen transports; (ii) its fourth cumulant is the exact diagonal plus an isotropic off-diagonal  $\lambda_l C^{\text{off}}$  with one fitted  $\lambda_l$  per layer, and it *feeds*  $\kappa_4$  into the fully off-diagonal  $\kappa_3$  (the largest single term beyond the base chain,  $2\times$  in MSE), whereas our transported  $(2, 2)$  slice is measured by them to be harmful off the diagonal ( $1.47\times$  worse than isotropic); (iii) the feedback coefficients of the GC arm are  $(\frac{1}{2}, \frac{1}{2})$  by the first-order Edgeworth count against our  $(1, \frac{1}{2})$ ; on network 0 changing ours to  $\frac{1}{2}$  moves the MSE from 5.74 to  $6.04 \cdot 10^{-7}$ , so this is not the bottleneck. The  $\rho$ -exact gates of (i) are the closed forms of Section 2: by Lemma 2.1 the coefficient of  $D21_{ac}$  in  $\kappa_3(h_a, h_a, h_c)$  is  $G_{ac} - \mu_a w_a \Phi((\alpha_c - \rho_{ac} \alpha_a) / \sqrt{1 - \rho_{ac}^2})$  with  $G_{ac} = \mathbb{E}[\theta_a \theta_c]$  the gate covariance (5), which reduces

to  $t_a \Phi_c$  at  $\rho = 0$ , and likewise for the transposed slice and the  $(2, 2)$  feed with conditional means and densities at  $z_c = 0$ . Every coefficient of the first-order transport is a Hermite series obtained by index shifts, which is how a system built on this note generates its term table (Section 9).

## 8 Reading: eight papers and the essence of the obstacle

The central obstacle of this note is quantitative. The exact first-order chain reaches  $3.5 \cdot 10^{-8}$  and, with the Euler–Stein constant,  $6.6\text{--}7.2 \cdot 10^{-9}$  on every network tried; what it costs is the memory of the third cumulant,  $12n^3$  per source-age and layer,  $0.5\text{--}0.7B$  in all, against a floor of  $0.1B$ . Section 7 shows that this memory is not compressible in the Frobenius sense (Hadamard obstruction), not sparsifiable in the hub index (flat importance), and not regenerable from the current layer (new shape at every layer); only the oldest ages collapse onto a shared basis. The eight papers below were read for what they say, between the lines, about an obstacle of this kind: a sum of many comparable terms whose readout is diagonal in a hidden index, needed to a relative precision of  $10^{-2}$ , under a budget linear in the width of the state.

### 8.1 What the papers say

*Hierarchical flows* [6]. A high-dimensional non-Gaussian probability is estimated and sampled through a hierarchy of conditionally Gaussian layers: at each scale the variance is a function of the *modulus* of the coarser scale, and the long-range non-Gaussian dependence of the field is carried entirely by interactions that are *local in the hierarchy*: a modulus at one scale coupled to the field at a neighbouring scale. The number of parameters is polylogarithmic in the dimension because the couplings are between hierarchy levels, not between coordinates. The robustness of the modulus ( $q = 1$  rather than the square) is a point the paper makes at length.

*Many Hamiltonians are sparsifiable* [7] and *optimal sparsifiers for Minkowski sums and sums of seminorms* [8]. A sum of  $m$  terms can be replaced by a reweighted subset of  $O(n/\varepsilon^2)$  terms that preserves the quadratic form, or the sum of seminorms, uniformly over all inputs in dimension  $n$ ; the proof is importance sampling under a non-redundancy condition, or, without logarithmic losses, the Reis–Rothvoss weight-perturbation scheme. Two statements matter here. The size of a sparsifier is governed by the *dimension of the input*, not by the number of terms, and it carries  $\varepsilon^{-2}$ . And there are sums that cannot be sparsified at all: sums of tensor-product (rank-one) terms of comparable weight, whose quadratic form is identity-like, need all their terms (Theorem 1.5 of [7]); random terms, by contrast, sparsify.

*Samplizer* [9]. Any quantum algorithm making  $Q$  queries to the reflection  $I - 2|\psi\rangle\langle\psi|$  about an unknown state can be run from  $\Theta(Q^2/\delta)$  copies of the state, and this quadratic overhead is optimal: the reflection, a rank-one perturbation of the identity, is built from samples by density-matrix exponentiation one small step at a time, each step with its fresh copy. The application estimates the angle between two states from the spectrum of the *product of two reflections* ( $e^{i(\pi \mp 2\gamma)}$ ) by phase estimation:  $\Theta(1/\varepsilon)$  queries,  $\Theta(1/\varepsilon^2)$  samples, both optimal. Coherent access to an object and sample access to it are separated by exactly a square.

*Adaptive phase estimation* [10]. Without a promise, adaptive (sequential) and non-adaptive (parallel) phase estimation need the same number of applications of the unitary; with a promise that the phase lies in a small known set, sequential strategies save a factor that is at most 2. The lower bound for non-adaptive strategies is a Farkas certificate: a nonnegative trigonometric polynomial exhibits the infeasibility of any initial state. Prior knowledge about the answer buys a constant, and the certificate of what it cannot buy is an explicit dual object.

*Efficient quantum Hermite transform* [11]. The harmonic-oscillator evolution factorises through  $\mathfrak{sl}_2$  (the number operator and the two ladder operators), which gives an exact fast-forwarding of the Gaussian (Ornstein–Uhlenbeck) semigroup: any time is one application, because the semigroup is diagonal in the Hermite basis with multiplier  $e^{-kt}$  on degree  $k$ . The same structure gives Hermite sampling and a Gaussian Goldreich–Levin algorithm that finds the heavy Hermite coefficients of a function from queries.

*Learning linear threshold functions under the Gaussian* [12]. The Hermite expansion of a ridge function is a symmetric tensor power of its direction,  $h_k(\langle w, x \rangle) = \sum_{|\alpha|=k} \sqrt{k!/\alpha!} w^\alpha h_\alpha(x)$ : the degree- $k$  Hermite tensor of a ridge function is rank one along  $w$ . A threshold function has Hermite weight  $\Theta(K^{-1/2})$  in every degree band  $[K, 2K]$ , and its gradient points along  $w$ , so  $O(\log(n/\varepsilon))$  gradient queries identify it, against polynomially many samples.

*Counting paths with an exterior algebra* [13]. A sum over exponentially many paths of products of per-vertex factors is estimated by a dynamic program whose state lives in a  $2^k$ -dimensional algebra (one exterior coordinate per subset of  $[k]$ , one auxiliary row of length  $D = k$  each): the identity  $g \wedge g = 0$  removes every walk with a repeated vertex exactly, independent Gaussian vectors make every surviving term contribute  $k!$  in expectation, and independent Gaussian matrices  $N(0, 1/D)$  attached to (position, vertex) pairs give path signatures whose products preserve second moments and whose fourth moment grows only like  $(1 + 2/D)^k$  (Lemma 6.3 there, from Rakhshan–Rabuseau). The matrices must be *shared* across all partial paths through the same vertex for the program to stay small; the moment bound is proved for exactly that reuse. Trees are counted by pinning a separator vertex and multiplying the component states. The relative second moment of one trial is  $O(k^2)$ , so  $O(k^2/\varepsilon^2)$  trials give relative error  $\varepsilon$ .

## 8.2 Four laws they hold for the obstacle

**(L1) The hub sum is in the non-sparsifiable class.** Each birth block is  $\sum_j a_j \otimes e_j \otimes c_j$ , one rank-one term per hub neuron  $j$  with the hub’s own coordinate vector as middle leg (Section 2), and at He initialisation the terms have comparable weight and are orthogonal in the slice inner product (the least-squares reweighting factor of the kept columns is 1.000). This is the identity-like sum of Theorem 1.5 of [7]: no reweighted subset reproduces it. The measurement agrees ( $q = \frac{1}{2}$ :  $\times 12$ ;  $q = \frac{1}{4}$ :  $\times 43$ ), and the sparsifier size  $O(n/\varepsilon^2)$  at the precision the chain needs,  $\varepsilon \approx 10^{-2}$ , exceeds the  $2n$  terms the block has. In the language of [1] the transverse measure of the hub foliation is uniform: there is no atom of dominant weight to keep.

**(L2) Nothing stochastic reaches the precision.** The memory enters the final mean through slices that must be right to about  $10^{-2}$  of themselves, and the slices are  $O(\rho) \approx 3\%$  residuals of  $O(1)$  objects. Any estimator that uses randomness to avoid the width of the state (probes for a Laplacian, sketches of the hub index, the exterior-algebra signatures) pays the sample law  $\varepsilon^{-2}$ : the sampler’s quadratic gap and the  $O(k^2/\varepsilon^2)$  trials of [13] are the same statement, and here it means  $10^4$ – $10^6$  probes against a budget of  $2n$  chain evaluations. This is the capacity obstruction of Section 3 again, now with its lower bound: coherent (exact) propagation costs the width of the state and no variance; sample access costs the square of the precision. For a trilinear readout the obstruction is sharper than for a bilinear one: a Gaussian sketch of the hub index has zero expectation on every trilinear form (odd moments vanish), so even an unbiased sketch needs third-moment-faithful signs (cube roots of unity) before the variance question arises.

**(L3) Reorganising the computation saves a constant at most.** The chain applies  $W_l$  to a state of width  $2n$  per age. The Gaussian weights are a promise about that state (its spectrum, its mean-field part), and [10] is the clean model of what a promise buys when the operator applications are the cost: a constant factor, never the order of growth, with a dual certificate for the rest. The identity-leg accounting and Strassen ( $24n^3 \rightarrow 12n^3 \rightarrow 10.5n^3$  per source-age) are that constant. The dual certificate here is the orthogonality of the hub terms in (L1).

**(L4) The crossover law of the hub-diagonal readout.** A source of age  $a$  has legs in the range of the cocycle  $T_{l \leftarrow l-a}$ . If that range is held in a basis  $Q$  of rank  $r$ , the transport of a source costs  $O(n^2 r)$  per layer instead of  $12n^3$ , but the slice  $\sum_j (QA)_{ij} (QB)_{ij} (QC)_{kj}$  is a Khatri–Rao contraction whose core costs  $\Theta(nr^3)$ : the hub-diagonal readout is cubic in the rank of the subspace. Hence a source is cheaper in its cocycle range only when

$$r^3 < n^2, \quad r < n^{2/3} \approx 101 \quad (n = 1024),$$

and the rank needed is the one at which the cocycle keeps the slice to  $10^{-2}$ , which by the Hadamard obstruction is the 99%-energy rank, not the stable rank. `scripts/diag_cocycle.py` gives, at layers 8–15, top-256 energy fractions 0.80, 0.92, 0.97, 0.99, 0.997, 0.998 at ages 1, 2, 3, 4, 6, 8: the 99%-rank is above 256 for ages 1–3 and below 100 from age five or six. The crossover law therefore predicts a dense window of four ages and a subspace for the rest, which is exactly the lossless configuration found by scanning (window 4,  $k = 384$ :  $3.69 \cdot 10^{-8}$ ) and not the cheaper ones (window 2:  $\times 2.2$ ). Its cost is  $4 \times 12n^3$  per layer before Strassen against a floor of  $12.8n^3$ : an exact first-order chain cannot be brought below 3–4 times the floor by any representation of its memory, because for the ages that carry the memory the dense Hadamard readout is already the cheapest one.

### 8.3 Three readings that point forward

**(P1) Fast-forwarding the Eldan path: the merge constant is a Hermite-degree average.** Eldan’s path is an Ornstein–Uhlenbeck semigroup in the tilt variable, and [11] is the reminder that such a semigroup is diagonal in Hermite degree: a defect whose kink content sits at degree  $d$  decays along the path like  $(1 + \tau)^{-d/2}$  (one kink: the density  $(1 + \tau)^{-1/2}$  of Section 3; two kinks: their product). Against the weight  $\frac{1}{2}(1 + \tau)^{-3/2}$  this gives, with the normalisation of Corollary 1.3,

$$a(d) = \frac{1}{d+1} : \quad a(1) = \frac{1}{2} \text{ (Gaussian closure: measured 0.535, 0.500)}, \quad a(2) = \frac{1}{3} \text{ (first-order chain: measured 0.500)}$$

the second reduced by the rotation of the defect direction along the path, as on the small network. The constant is therefore a property of the *order* of the chain, not of the network, which is what the cross-over of  $a_0$  and  $a_1$  without loss on both networks shows. A system can fix it offline from the chain’s order and never fit it.

**(P2) The memory is a sum over hub-rooted stars, and one of its three legs collapses.** With  $S_{l'} \approx T_{l' \leftarrow 0} T_{l' \leftarrow 0}^\top$  the star block born at  $l'$  and read at  $l$  is

$$\sum_j (T_{l \leftarrow l'} \Phi S_{l'})_{ij} (T_{l \leftarrow l'})_{ij} (T_{l \leftarrow l'} w_2 S_{l'} \Phi)_{kj} = \sum_j \sum_{m, m'} T_{l \leftarrow l'}(i, j) T_{l' \leftarrow 0}(j, m) T_{l' \leftarrow 0}(j, m') A_{im}^{(l')} B_{km'}^{(l')},$$

a sum over hub vertices  $j$  at every layer  $l'$  of two paths from the input to the hub and one path from the hub to the output, the tree-counting situation of [13] with the hub as separator. The first

leg accumulates:  $U_l = \sum_{l' \leq l} T_{l \leftarrow l'} D(\Phi_{l'}) T_{l' \leftarrow 0}$  obeys  $U_{l+1} = W_{l+1} D(g_{l+1}) U_l + D(\Phi_{l+1}) T_{l+1 \leftarrow 0}$ , one  $n \times n$  matrix for all sources at  $n^3$  per layer. What does not collapse is the pairing of the hub leg  $T_{l \leftarrow l'}(i, j)$  with the third leg at the same  $(l', j)$ : the Hadamard pairing is diagonal in the hub and the hub's age is a label the sum cannot forget. This locates the memory precisely: not in the transported content (which accumulates) but in the hub-diagonal pairing of the identity leg with the readout leg of the same birth. The coherent part of that pairing is the mean of  $T \circ T$  over the weights, a variance profile computable at  $O(n^2)$ , and the rank-one experiments of Section 7 measure it to carry 80% of the memory; the rest is the realisation-specific fluctuation of a Gaussian matrix product, of the relative size the fourth-moment bound of [13] predicts for products of  $a$  matrices of width  $n$ , a few percent per age.

**(P3) One object: memory, tilt response, defect.** Theorem 2.2 writes the Euler–Stein defect as  $-\langle \nabla^2 G, \Gamma(y) \rangle$  with  $\Gamma$  built from the first-order tilt responses, and the second-order response of the Gaussian chain to the  $n$  coordinate tilts has, at every layer, the birth structure  $\sum_a (\text{cocycle column}) \otimes e_a \otimes (\text{row of } X_l)$ : an identity leg born at each layer and transported by the same cocycle. The third cumulant's memory, the tilt-response tensor and the defect are the same cocycle-transported identity-leg object read through the same hub-diagonal pairing. Two consequences. The symbolic Euler–Stein correction (the open task of Section 7) costs what the memory costs, so it is not an add-on to the exact chain but a second use of the same transported legs, to be built on them. And the cheap part of each is the same cheap part: the mean-field pairing at  $O(n^2)$ .

#### 8.4 What this says the system is

An exact first-order chain cannot reach the floor: (L4) puts it at three to four times  $0.1B$  after every algebraic saving, and (L1)–(L3) close the other exits. A floor-level estimator with the accuracy of the leaders is therefore not an exact chain with a compressed memory; it is a chain with a *short* memory, corrected by derived quantities that cost  $O(n^2)$  per layer: the mean-field hub pairing for every age beyond the window (P2), the radial scalar of Lemma 4.1 (one number, worth a factor 16), and the Euler–Stein merge with the constant fixed by the order of the chain (P1), computed from the same legs as the memory (P3). The one measurement that decides whether this reaches  $10^{-8}$  at the floor is whether the defect of a *short-memory* chain is as explainable by its own coordinate Laplacian as that of the Gaussian closure (89–93%) and of the exact chain (81–84%): if it is, a window-one chain at the floor with the mean-field tail and the merge lands near  $10^{-8}$ ; if it is not, the memory is the price and the score is bounded below by the crossover law. That measurement is 2049 runs of the window-one chain, which the Modal batch of Section 7 does in an hour; it is the next thing to run.

## 9 Dictionary and open problems

### 9.1 Dictionary

| stage 8 (commutant, modular tilts, leaf space)              | stage 9 (MLP mean estimation)                                                                                                                                                                                               |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| state $\nu$ on the algebra; affine pairing $\nu(F)$         | input law; the exact answer $u(\nu) = \nu(h_L)$                                                                                                                                                                             |
| localization process (positive martingale in the commutant) | Eldan tilt of the input, directional tilt, pinning; on the neuron side, the layer-by-layer closure                                                                                                                          |
| driving covariance $C_t$ ; trickle-down term $KC_tK$        | which errors a scheme sees: the second variation $E''$ against $d\langle\nu\rangle$ (Theorem 1.2)                                                                                                                           |
| modular tilt of a Gaussian state                            | shift of the mean; its action on the chaos expansion is the index shift (Lemma 2.1)                                                                                                                                         |
| central (trivial) part of the localization                  | the collective activity mode $s = \mathbb{E}z^2$ , a scale mixture (Lemma 4.1); the transverse measure is its law                                                                                                           |
| Cartan part                                                 | the bulk of the covariance, harmless at percent accuracy (Proposition 4.2)                                                                                                                                                  |
| Kikuchi–Toeplitz quantization                               | the heat semigroup in the input mean (Price’s theorem); the exact answer is the Berezin transform of the network function, the chain a Toeplitz symbol that is not heat-harmonic, and the anomaly is the Euler–Stein defect |
| capacity of a localization                                  | Proposition 3.1: input tilts have signal-to-noise $\sqrt{k} \delta /\ A\ _F \approx 10^{-2}\sqrt{k}$                                                                                                                        |
| births / creation operators                                 | the kink births $w_a = \varphi(\alpha_a)/\sigma_a$ , second derivatives of the tilt response ((3))                                                                                                                          |

### 9.2 What the measurements prescribe for a system, and what is open

The decisive constraints are the pair floor of Proposition 6.1 ( $12.8n^3$  per layer at the  $0.1B$  floor), the price of one source-age of exact first-order memory ( $12n^3$ , Section 6), and the error accounting of Section 7: the central variable is one scalar per layer and worth a factor 16; the memory of  $\kappa_3$  is worth a factor 100, is smooth in amplitude and brittle in shape; a window of four ages with a  $k = 384$  tier is lossless and costs about  $0.5B$  with every structural saving; the leaders sit at  $0.1B$  with three times the accuracy of the exact first-order chain. In this order of expected value:

1. **Memory, not regeneration.** The shape diagnostic of Section 7 rules out the naive version of the following, so it is stated as the open problem it is: The old sources’ effect is smooth in amplitude (10% scale: factor 1.5) and their transport is the collective mode; the open-source chain already regenerates the off-diagonal  $\kappa_4$  from the covariance with one fitted  $\lambda_l$  per layer. The analogous move for  $\kappa_3$  would be to keep one or two dense ages and replace the rest by a function of the current state (the gated covariance and the current slices) with per-layer fitted scalars, fitted on the public networks against the dense chain’s slices rather than against the final means. The theory says what the function should be: the old content is the image of births under the cocycle, i.e. Hadamard products of transported covariances, so the ansatz is the star block of the present layer with its own gates scaled by the cocycle’s decay.
2. **Euler–Stein merge.**  $e - a\delta(0)$  with  $a$  fitted once: a factor 9 for the Gaussian closure at  $n = 1024$  ( $a = 0.54$ , midpoint  $a = \frac{1}{2}$  nearly as good), measured for the exact first-order chain in Section 7. Affordable only through the closable part of the second-order response of Theorem 2.2, whose five objects are the same as a  $\kappa_3$  chain’s; whether it correlates with the exact  $\delta$  as well as  $\delta$  with the error is the open experiment.

3. **Identity-leg structure and Strassen** for whatever dense memory is kept:  $12n^3$  per source-age, 8.3 with Strassen, against the 24 of the reference code.
4. **Noncommutative version.** The gates  $\theta(z_a)$  of one layer generate a commutative algebra; across layers the gate algebras do not commute with the weights, and the chain’s closure is a product state on the gate algebra of layer  $l$  conditioned on the surrogate of layer  $l - 1$ . The stage-8 KMS condition for the modular flow of the Gaussian state, restricted to the gate algebra, is the heat equation of Section 2; its failure across layers is the defect. Writing the chain as a state on the free product of the gate algebras, with the trickle-down of [3] as its Cartan expansion, is the theoretical continuation; the cocycle of Section 6 is its transverse measure.

## A Verification

`scripts/verify_stage9.py` checks, in order: (1) Lemma 2.1 to  $2 \cdot 10^{-8}$  for  $k \leq 10$ ; (2) Price’s theorem and Stein’s lemma for the bivariate RELU moment,  $\partial_{m_a} M_{ab} = \mathbb{E}[\theta_a \text{relu } z_b]$ ,  $\partial_{s_{ab}} M_{ab} = \mathbb{E}[\theta_a \theta_b] = \partial_{m_a} \partial_{m_b} M_{ab}$ , against  $4 \cdot 10^6$  Monte Carlo samples; (3) Theorem 2.2 on a network with  $n = 6$ ,  $L = 3$ : the gradients of the first-layer moments in closed form against finite differences ( $1.6 \cdot 10^{-7}$  on a scale of 2), and  $\delta(0)$  by finite differences in the input mean against  $-\langle \nabla^2 \mathcal{G}, \Gamma \rangle$  with the Hessian of the downstream chain by finite differences in the 27 moment coordinates: agreement to  $7 \cdot 10^{-7}$  on  $|\delta|_{\text{rms}} = 0.28$ ; (4) Lemma 4.1 by Gauss–Hermite quadrature in the scale,  $v = 10^{-2}$ : the closure error matches  $-\frac{v}{2}\sigma(\alpha^2 - 1)\varphi$  and the  $\kappa_4$  readout cancels it to the  $O(v^2)$  level for  $\alpha \in \{-1, 0, 0.7, 1.5\}$ . `scripts/defect_small.py`, `lap_exact.py`, `lapx_merge.py` produce the numbers of Sections 3 and 7; `diag_eig.v3.py`, `diag_onestep.py`, `injected.py` those of Section 4.

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